
ROTATING LED DISPLAY

Wireless Control & Persistence of Vision (POV) Platform

ATmega32-Based Embedded Systems Design

Student IDs: 2205139, 2205143, 2205147

Project Proposal & Technical Architecture

1. Motivation

Problem Description & Importance

Problem Description

- Traditional flat electronic displays (LCD/OLED/LED matrices) require large physical footprints, heavy mounting frames, and high component counts.
- Viewing angle limitations restrict visibility in 360° public environments or cylindrical installations.
- Displaying dynamic text and real-time graphics on a rotating body creates complex synchronization challenges due to mechanical jitter and phase drift.

Why It Matters

- Persistence of Vision (POV) technology exploits human visual latency (≈ 100 ms retention) to render high-resolution 360° displays using only a single spinning array of LEDs.
- Drastically reduces cost, weight, power, and component count compared to multi-panel display grids.
- Demonstrates real-time microsecond-level embedded timing, precise external interrupt synchronization, and multi-protocol wireless integration.

1. Motivation

Why This Project & Main Objectives

The project combines high-speed motor rotation, microsecond-accurate LED blinking, real-time wireless Bluetooth communication, RTC timekeeping, and interrupt-driven phase locking on the ATmega32 microcontroller to build a floating cylindrical POV display.

1. **High-Resolution POV Rendering:** Blink a 24-LED array with microsecond precision during rotation to generate crisp alphanumeric characters, digital clocks, and analog clock hands without visual flicker.
2. **Interrupt-Based Phase Synchronization:** Interface an IR obstacle sensor with external interrupt hardware (INT0) to detect rotation cycle completion and eliminate display drift/rotation.
3. **Real-Time Clock Synthesis:** Integrate a DS3231 RTC module over I²C (TWI) to display continuous analog (hour/minute/second hands across 60 angular ticks) and digital clocks.
4. **Wireless Dynamic Telemetry:** Receive custom text strings over Bluetooth (HC-05 hardware USART) and render them on-the-fly without interrupting rotation phase locking.
5. **Onboard Power Autonomy:** Implement a 3.7 V Li-Ion battery power circuit with onboard charging to eliminate trailing wires on the spinning arm.

1. Motivation

Real-World Deployment Scenarios

Operational Sector	Technical Challenge	POV Platform Solution
360° Cylindrical Public Advertising Compact Public Information Kiosks Industrial Event Telemetry	Multi-panel displays have dead spots and high power draw Limited enclosure space for bulky multi-digit LED displays Cable entanglement on rotating machinery	Single spinning LED arm renders seamless 360° cylindrical text/logos with 90% power savings. High-speed rotating PCB projects floating digital/analog clocks and live notifications in a compact footprint. Onboard Li-Ion battery power + wireless HC-05 Bluetooth serial link streams real-time messages cable-free.

All scenarios run on the same core interrupt-driven FSM – only incoming Bluetooth commands and display modes differ.

- **Visual Efficiency:** Uses visual memory to transform a 1D line of 24 LEDs into a 2D cylindrical surface display.
- **Dynamic Remote Reconfiguration:** Bluetooth wireless interface allows operators to update displayed text instantly from any mobile terminal.
- **Hardware Integrity:** Self-contained battery power supply avoids slip-ring noise and mechanical brush wear.

2. Equipment

Hardware Components & Justification

Component	Specification / Count	Purpose & Justification
Microcontroller	ATmega32-16PU (DIP-40) ×1	Core system controller. Provides 32 GPIO lines for 24-LED direct port control, hardware INT0 for IR phase anchor, USART for HC-05, and TWI for DS3231 RTC.
LED Array	High-Brightness 5mm LEDs ×24	Primary visual display element mounted on the spinning panel; lit in bit-pattern columns to form text and clock hands.
Phase Anchor Sensor	IR Obstacle Sensor Module ×1	Positioned at fixed frame anchor; triggers external interrupt (INT0) once per rotation to reset phase counter and eliminate image drift.
Wireless Module	HC-05 Bluetooth Module ×1	Interfaces with hardware USART (PD0/PD1) to receive custom text strings wirelessly from smartphones or PCs.
Real-Time Clock	DS3231 RTC Module ×1	High-precision I ² C RTC module; provides real-time timekeeping for analog clock (60 angular slots) and digital clock display.
Drive Motor	High-RPM DC Motor ×1	Rotates the LED panel at stable RPM (≥ 1000 RPM) to achieve visual persistence above 15–20 Hz frame rate.
Power Circuit	3.7V Li-Ion + Charging Module ×1	Onboard battery mounted directly on spinning PCB, providing wireless power to LEDs and MCU without heavy slip rings.
Stationary Power	5V Wall Charger ×1	Bench power supply for motor drive unit and stationary battery charger circuit.

2. Equipment

Microcontroller Evaluation: Arduino Nano vs. ATmega32

Metric / Feature	Arduino Nano (ATmega328P)	ATmega32-16PU (Selected)	Justification & Feature Preserving Verdict
GPIO Count	22 Digital/Analog Pins	32 Dedicated I/O Lines	ATmega32 provides 4 full 8-bit ports (PA, PB, PC, PD). Direct 24-LED driving without external multiplexer shift registers.
Port Writing Speed	Multi-pin bit shifting required	Single-cycle 8-bit PORT write	Writing PORTA, PORTB, PORTC in 1 clock cycle gives ultra-sharp LED column switching for high-RPM spinning.
UART Baud Precision	Subject to clone bootloader baud bug (38400 → 9600 mismatch)	0.0% Baud Error with 11.0592 MHz crystal	Eliminates the market-clone Arduino Nano hardware baud error reported in early testing; ensures reliable 115200/38400 Bluetooth transfers.
Hardware Interfaces	1× SPI, 1× I ² C, 1× USART, 2× Ext Int	1× SPI, 1× TWI, 1× USART, 3× Ext Int	Retains 100% functionality (USART for HC-05, TWI for DS3231, INT0 for IR sensor) while offering 10 extra GPIO pins and native 40-pin socketing.
Is Nano Mandatory?	Convenience dev-board	NOT strictly mandatory	Moving to ATmega32 retains 100% features, improves execution timing, and satisfies academic embedded system criteria.

Conclusion: Arduino Nano is NOT strictly mandatory. Moving to ATmega32 preserves all capabilities, fixes serial baud mismatches, and simplifies 24-LED driving via full 8-bit port registers.

2. Equipment

Software & Development Tools

Toolchain & Compilers

- **Microchip Studio / AVR-GCC** – Bare-metal C development, register-level drivers, and hex generation.
- **USBasp Programmer & eXtreme Burner** – Direct ISP flashing into ATmega32 Flash memory.
- **Proteus VSM** – Full circuit schematic design and timing simulation before PCB fabrication.

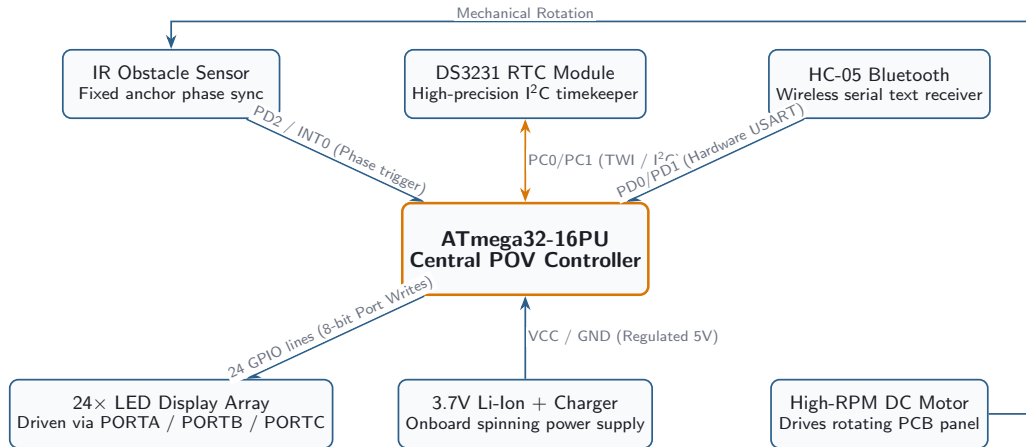
The ATmega32 handles phase detection, timing delays, bit pattern output, and RTC reads autonomously. The Bluetooth terminal is strictly an input medium.

Host & Testing Software

- **Serial Bluetooth Terminal (Android / PC)** – Wireless transmission of custom ASCII strings to HC-05.
- **Custom Bit-Matrix Generator** – Python/C utility to map ASCII characters into 8×5 and 24×5 LED column matrices.

3. System Design

System Hardware Block Diagram



→ Signal / Control / Mechanical path ↔ Bidirectional I²C communication path.

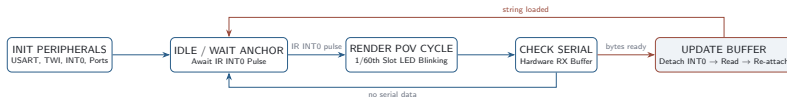
3. System Design

ATmega32 Interface Pinout Allocation Map

Pin(s)	Function	Connected Component & Operational Role
PA0 – PA7	GPIO Port A Output	LED Pins 1–8 (Lower 8 LEDs of 24-LED array); single-cycle PORTA write
PB0 – PB7	GPIO Port B Output	LED Pins 9–16 (Middle 8 LEDs of 24-LED array); single-cycle PORTB write
PC2 – PC7, PD6, PD7	GPIO Port C/D Output	LED Pins 17–24 (Upper 8 LEDs of 24-LED array); single-cycle PORTC/D write
PD2 (INT0)	Hardware External Interrupt 0	IR Obstacle Sensor digital output; triggers on anchor pulse to synchronize cycle phase
PD0 (RXD), PD1 (TXD)	Hardware USART	HC-05 Bluetooth module at 38400 / 9600 baud for wireless text reception
PC0 (SCL), PC1 (SDA)	Hardware TWI (I ² C)	DS3231 RTC Module for time data retrieval (7-bit address 0x68)
XTAL1, XTAL2	Clock Oscillator	11.0592 MHz crystal + 22 pF ceramic caps for exact UART baud generation (0% error)
RESET	Master Reset Pin	10 kΩ pull-up + tactile push button for manual system reset
VCC, GND, AVCC	Power Lines	Regulated 5V supply from onboard 3.7V Li-Ion boost converter circuit

4. Role of the ATmega32

Firmware Execution Flow & Interrupt Management



- **Interrupt-Driven Rotation Lock:** The IR sensor output triggers an external interrupt on INT0 at the anchor position, marking the exact 0° baseline of each rotation cycle.
- **Angular Division (1/60th Slots):** The ATmega32 divides the measured rotation period into 60 equal angular increments to render 60-position analog clock hands (hour, minute, second) and aligned text characters.
- **Bit-Matrix Column Driver:** Writes column bit-patterns directly to GPIO ports with calibrated delay routines to achieve optimal letter width and spacing without blurring.
- **Serial Collision Decoupling:** When Bluetooth data arrives, INT0 is temporarily detached during the UART string buffer transfer and re-attached immediately after, preventing serial interrupt collisions from corrupting the display cycle.

4. Role of the ATmega32

Peripheral Utilization Breakdown

- **GPIO Port Register Control (PA, PB, PC, PD):** Direct 8-bit parallel writes to drive 24 LEDs simultaneously. Executing single-cycle PORT assignments guarantees microsecond-level LED column switching essential for high-RPM display sharpness.
- **External Interrupt 0 (INT0 / PD2):** Falling-edge trigger from the IR obstacle sensor. Captures rotation phase start instantly, resetting the column index counter and eliminating image rotation drift.
- **Hardware TWI / I²C (PC0 / PC1):** Operates at 100 kHz to communicate with the DS3231 RTC module. Fetches seconds, minutes, and hours data during idle phase gaps.
- **Hardware USART (PD0 / PD1):** Configured for 38400 / 9600 baud wireless serial data reception from HC-05. Uses double-buffered RX interrupts to buffer incoming ASCII strings.
- **Timer0 / Timer1 (16-bit / 8-bit):** Provides precision microsecond timebase for calculating rotation period (T_{rot}), column delay times ($\Delta t_{col} = T_{rot}/N_{cols}$), and letter spacing intervals.

5. Implementation Plan

Development Stages & Milestones

Project Update 1 (Wks 1–3)

- 1.1 Bare-metal ATmega32 clock, GPIO ports, and timer setup.
- 1.2 IR obstacle sensor integration on INT0 for period measurement.
- 1.3 LED column bit-pattern rendering for single static characters ('S', 'A', digits).
- 1.4 Mechanical motor balancing and basic rotating panel assembly.

Deliverable: Rotating panel displays a steady single character locked to the IR anchor position without drift.

Final Demonstration (Wks 4–6)

- 2.1 Hardware TWI driver for DS3231 RTC module time sync.
- 2.2 Hardware USART integration for HC-05 Bluetooth dynamic text input.
- 2.3 Implementation of INT0 detach/re-attach serial decoupling routine.
- 2.4 Rendering routines for Analog Clock (60 slots), Digital Clock, and Scrolling Text.
- 2.5 Onboard Li-Ion power circuit integration and final PCB enclosures.

Deliverable: Complete wireless floating POV display showing live analog/digital clock and Bluetooth text messages.

5. Implementation Plan

Group Member Responsibilities

Member 1: 2205139

- Firmware Architecture & Core Timing Logic
- INT0 interrupt anchor handler & 1/60th angular slot calculation
- INT0 detach/re-attach serial decoupling logic

Member 2: 2205143

- Hardware, Motor Drive & Power Systems
- 24-LED rotating PCB fabrication & motor mounting
- 3.7V Li-Ion battery power regulation & mechanical balancing

Member 3: 2205147

- Peripheral Interfaces & Software Utility
- DS3231 RTC TWI driver & HC-05 USART driver
- Character bit-matrix encoding & mobile Bluetooth terminal testing

6. Use of Other Controllers

Controller Selection & Arduino Nano Replacement Justification

Main System Architecture

The **ATmega32-16PU** serves as the sole central controller for the entire project. It directly manages the IR phase anchor interrupt, DS3231 RTC TWI bus, HC-05 Bluetooth USART communication, and 24-LED array port drivers.

Evaluation of Secondary / External Controllers

- **Why Arduino Nano was initially considered:** Popular dev-board choice in early prototyping due to integrated USB-to-serial chip and ready-made header pins.
- **Is Arduino Nano strictly mandatory? NO.** It is an ATmega328P breakout board. Replacing it with the ATmega32 results in **zero feature loss** and improves hardware design.
- **Why ATmega32 is Superior for this Project:**
 - **Direct 8-Bit Port Manipulation:** ATmega32 has 32 I/O lines across 4 clean 8-bit ports (PA, PB, PC, PD). All 24 LEDs connect directly to full ports, enabling instant 1-cycle PORT writes for sharp LED column switching.
 - **Fixes Clone Nano Baud Rate Bug:** Eliminates the hardware baud rate mismatch observed on cloned Nano boards (where 38400 baud operated at 9600 due to bad fuses/crystals). Using ATmega32 with an 11.0592 MHz crystal gives 0.0% baud error.
 - **No Auxiliary Controllers Required:** No ESP32 or Raspberry Pi is needed. ATmega32 alone possesses sufficient memory, timers, and hardware peripherals to execute all tasks autonomously.

7. Testing & Success Criteria

Measurable Criteria & Test Methodology

Metric	Target Value	Test Method & Validation Tool	Priority
Phase Sync Stability	0° image rotation drift	Visual inspection under stroboscopic/steady rotation over 10 minutes	Essential
Frame Refresh Rate	≥ 20 Hz (≥ 1200 RPM)	DC motor tachometer check; ensures flicker-free human vision retention	Essential
Interrupt Latency	$< 5 \mu\text{s}$ on INT0	Oscilloscope check from IR sensor pulse edge to ISR execution start	Essential
LED Column Switch Speed	$< 1 \mu\text{s}$ per column	Direct PORT write execution check on logic analyzer	Essential
Bluetooth Transmission	100% ASCII string integrity	Send 50-character strings via Serial Bluetooth Terminal at 38400 baud	Essential
RTC Time Synchronization	± 1 second precision	Compare displayed analog/digital clock hands against DS3231 hardware RTC	Essential
Serial Decoupling Safety	Zero display glitching during RX	Send continuous Bluetooth text streams while POV rendering is active	Essential
Power Battery Autonomy	≥ 45 minutes continuous	3.7V Li-Ion discharge test under full 24-LED spinning load	Essential
Wireless Control GUI	Smartphone app interface	Optional custom Android app with buttons for mode switching	Optional

Essential features: INT0 phase locking, 24-LED PORT driving, DS3231 RTC clock sync, HC-05 USART text reception, and serial decoupling. Optional features: Custom smartphone UI app.

8. Challenges & Future Scope

Anticipated Technical Issues & Scaling Path

Technical Challenges & Solutions

- **Bluetooth Baud Rate Mismatch:** Cloned Arduino Nanos showed baud rate errors (38400 settings acting as 9600) → **Solution:** Switch to ATmega32 with dedicated 11.0592 MHz crystal for exact 0.0% UART baud calculation.
- **Sensor Jitter (Hall vs IR):** Hall magnetic sensor signals proved jittery and imprecise → **Solution:** Replaced with an IR obstacle sensor, providing clean digital pulses for INT0 anchor locking.
- **Timing vs Interrupt Drift:** Periodic timing alone caused continuous image rotation → **Solution:** Re-synchronized display loop on every rotation cycle using INT0 falling-edge interrupt.
- **Serial-Display Collision:** Receiving Bluetooth data inside ISR caused display overlap → **Solution:** Temporarily detach INT0 interrupt during UART string reading, then re-attach immediately after buffer update.
- **Letter Spacing & Column Timing:** Blur occurs if LED ON-time is improperly calibrated → **Solution:** Microsecond-level timer delay calibration based on real-time T_{rot} .

Future Scope

Near-Term Enhancements

- **Wireless Power Transfer / Slip Rings:** Implement inductive wireless power coils on the motor base to eliminate Li-Ion battery recharging downtime.
- **RGB Full-Color LEDs:** Upgrade 24 mono LEDs to WS2812B / APA102 RGB LEDs to render full-color graphics and animations.

Long-Term Scaling

- **3D Holographic Display:** Stagger multiple rotating vertical LED arcs at offset angles to create full 3D volumetric holographic projections.
- **Dedicated Mobile App GUI:** Build a custom Android/iOS app with visual drawing canvas to stream real-time doodles directly to the display over Bluetooth.